HipMRI segmentation: midpoint feasibility review

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1. Requirements and scope

For MRI researchers, build a reproducible 2D multiclass segmentation prototype and determine whether an Improved U-Net reduces boundary errors enough to justify its compute cost. This is not a clinical deployment. **Proposed final acceptance criteria**, not achieved results:

1. Zero patient overlap across train/validation/test; verified image-mask pairing, one global label dictionary and a confirmed prostate class.
2. Mean patient-level prostate Dice >= 0.75 on the held-out test set; also report per-class Dice/IoU and variation across patients.
3. Improved U-Net gains >= 0.02 prostate boundary F1 over the baseline at a fixed two-pixel tolerance, with prostate Dice decreasing by no more than 0.01.
4. At the chosen real-data configuration, peak PyTorch allocated training memory <= 6 GiB and median batch-one model inference <= 100 ms on the local GPU.
5. Record seeds, splits, configuration and checkpoint reload agreement; explain 3-5 genuine failure cases. An unmet accuracy target remains a reportable finding.

2. Model, data and rationale

Use course HipMRI slices at `/home/groups/comp3710/HipMRI_Study_open/keras_slices_data`. Baseline: Ronneberger et al.'s 2D U-Net [1]. Main model: the **Normal-difficulty** 2D Improved U-Net option, adapting Isensee et al. [2] with residual context blocks, instance normalization, spatial dropout and deep supervision. This applies multiscale CNN features, skip connections, regularization and gradient propagation. It is a configurable 2D adaptation, not a complete reproduction of the paper's 3D BraTS training or ensemble. Compare identical patient splits, preprocessing and training budgets using cross-entropy plus foreground Dice loss. Preserve all verified classes; do not merge every nonzero label into prostate.

3. Preliminary feasibility evidence

Both models completed six epochs on **synthetic geometric data**: 32 training and 8 validation images, 128 x 128, seed 3710. Saved/reloaded predictions agreed exactly. Tests cover patient separation, label validation, model gradients and metric edge cases. These are pipeline checks, not medical accuracy evidence.

Local RTX 4070 Laptop GPU (8 GB), float32, base width 16: a separate short tensor probe used 256 x 256 inputs, training batch 4, **two output classes**:

Measurement	U-Net	Improved U-Net
Parameters	1,940,834	3,186,342
Peak allocated / reserved memory (GiB)	0.290 / 0.438	0.639 / 0.820
Median model inference, batch 1 (ms)	1.395	4.224

Sources: `results/resource_*.json` and synthetic summaries. CUDA timings are synchronized; inference excludes preprocessing/I/O. Allocator memory excludes driver/other processes. Actual HipMRI multiclass costs must be remeasured. **Real data access, patient metadata and label meanings remain unverified.**

4. Risks, budget and next experiments

Provisional post-midpoint budget: 16 working hours and 12 GPU-hours, revised after a real-data pilot. Next: verify Rangpur access; audit labels, patient IDs, spacing and overlays; freeze patient splits; run a small U-Net pilot; then compare models on validation data. Keep test data locked until model choices are finalized. Risks are delayed access, mislabeled classes, leakage, limited memory and poor apex/base boundaries. If access is delayed, present measured engineering evidence and the outstanding dependency. If memory/time is insufficient, reduce batch/width for both models and document the change; retain U-Net as the fallback. Tutor discussion and personal check-off remain required.

References: [1] Ronneberger et al., 2015. [2] Isensee et al., 2018. Assessment: `COMP3710_Report_2026_draft_v2.2.pdf`, Sections 2.2 and 3.